

RESEARCH SEMINAR BY ELLA GOODBINDER

SOUND THERAPY FOR SLEEP



INTRODUCTION

The purpose of this seminar is to discuss the different types of sound therapy that have been studied as treatments for insomnia and sleep disorders. Specifically, we will be looking at studies that discuss how different sounds and frequencies decrease sleep latency. The most studied audios involve white or pink noise, but there is also speculation about the effects of binaural beats and audio at 432 Hertz.



PURPOSE

Although sound therapy is used to treat sleep disorders, it's overall effectiveness is still up for debate. I am connecting my research to my final project in this class by allowing you all to decide for yourself if you find different audio tracks effective at making you sleepy or not.

I do not expect any participate to fall asleep in the five minutes in my experience, but I do think you would definitely feel relaxing effects after a couple minutes.



IMPORTANT DEFINITIONS:

Hertz: The frequency of a sound wave, determines its pitch



Sleep Latency

- The time it takes to transition from full wakefulness to sleep when laying in the dark with closed eyes.
- The average healthy person's sleep latency is 10-20 min.
- Sleep Latency decreasing means falling asleep faster, and increasing it means it takes longer to fall asleep.



Binaural Beats

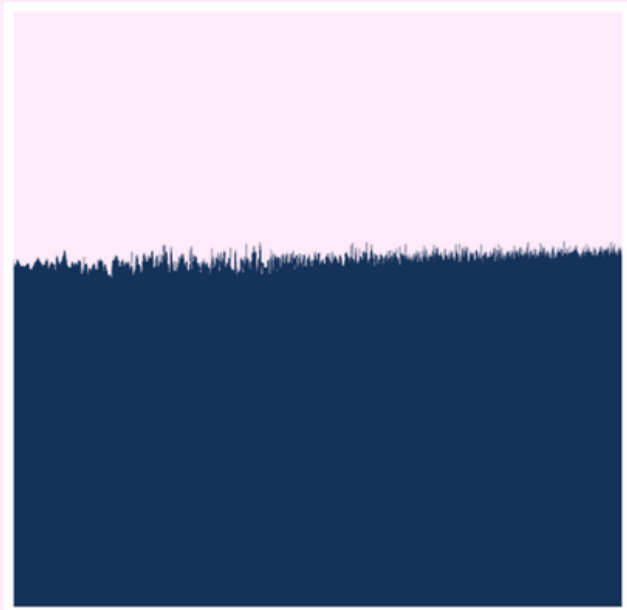
- When you listen to two slightly different tones in each ear, your brain perceives a third tone at the frequency difference between the two sounds.
- To experience these you need headphones to play one tone in each ear.
- Ex: 530 Hz tone in one ear and a 520 Hz tone in the other will create the illusion of a 10 Hz beat.



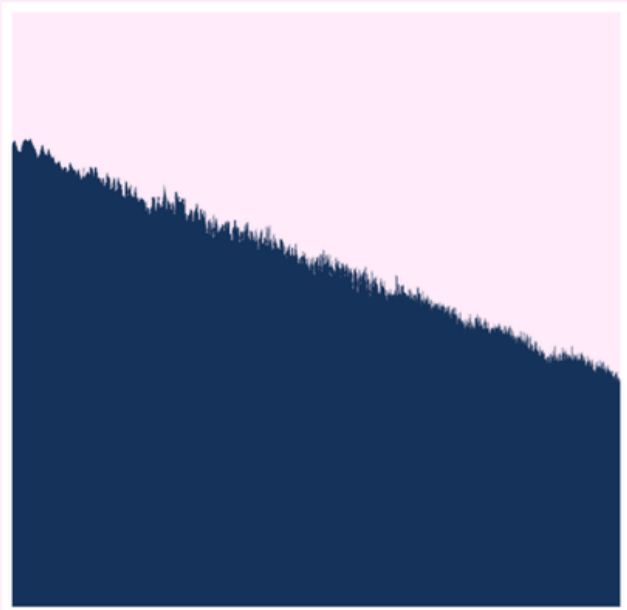
White/Pink Noise

- White noise results from combining sounds of all different frequencies so that the intensity is the same at every frequency.
- It is like white light, which contains all visible color frequencies.
- Pink noise also contains all frequencies that the human ear can hear, but the intensity of the sound decreases as frequency increases. (Natural sound)

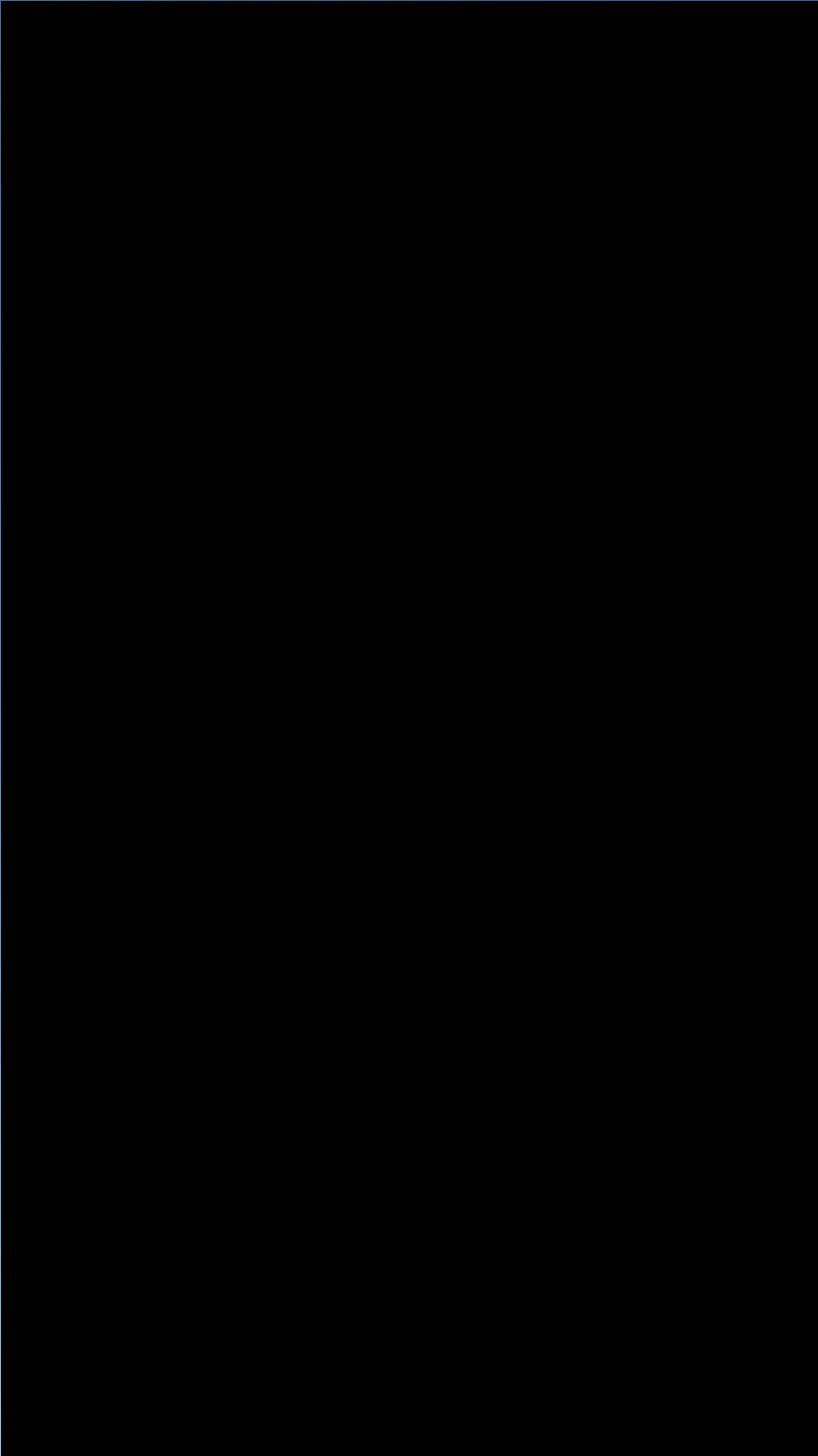
White Noise Vs Pink Noise



White Noise



Pink Noise



WHITE/PINK NOISE

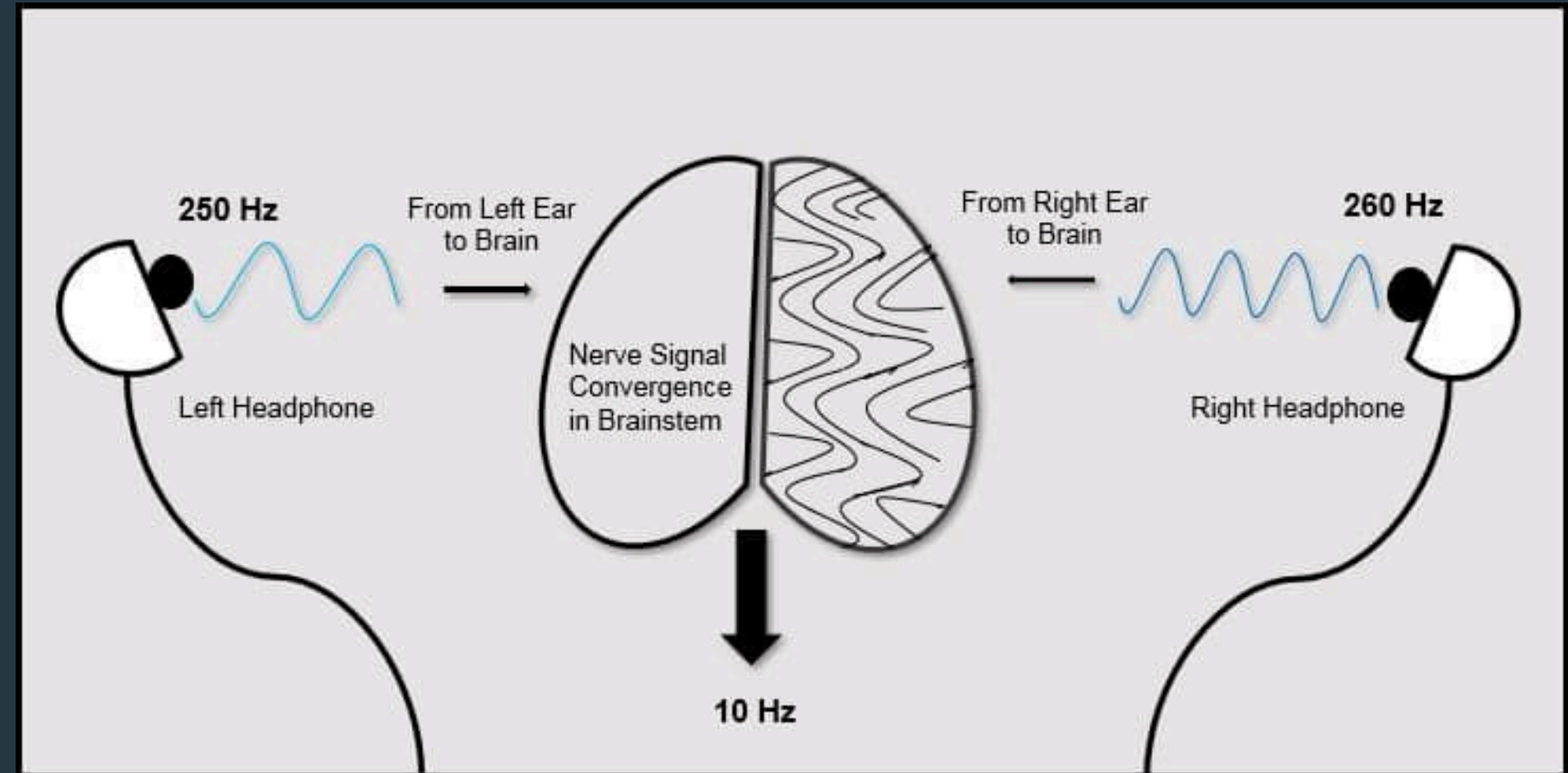
- Of the audios to sleep to, white noise is the most studied, and it seems to have positive sleep effects in some scenarios.
- Sleep disturbances are more likely to happen with a changing auditory environment that is not constant, and white noise can play a part in evening out a noisy environment to promote better sleep quality and sleep latency.
- White noise was only concluded to have a positive effect in 1/3 of studies
- Pink noise was concluded to have a positive effect in 80% of them



BINAURAL BEATS

- There have been studies conducted that show positive effects on sleep latency.
- Just like white noise it is still not officially “proven” to be beneficial. There just is not much research done

There are a couple of accounts like this one that really love posting music videos to sleep to, not all are actually binaural beats. It certainly sounds fancy enough to be clickbait though ---->

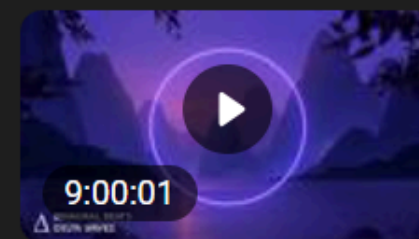


Videos :



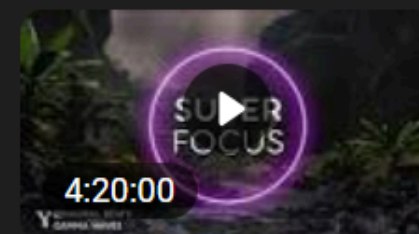
[SUPER FOCUS \[Flow State Music \] Binaural Beats 40Hz ...](#)

YouTube · SleepTube - Hypnotic Relaxation ·
Oct 5, 2022



[\[SLEEP 8 HOURS\] "Indigo Dreamstate" | Delta 3Hz Binaural ...](#)

YouTube · SleepTube - Hypnotic Relaxation ·
3 weeks ago



[40 Hz Study Music - GAMMA Binaural Beats, Brainwave Music ...](#)

YouTube · SleepTube - Hypnotic Relaxation ·
May 23, 2025

432/440 HERTZ AND OTHER FREQUENCIES

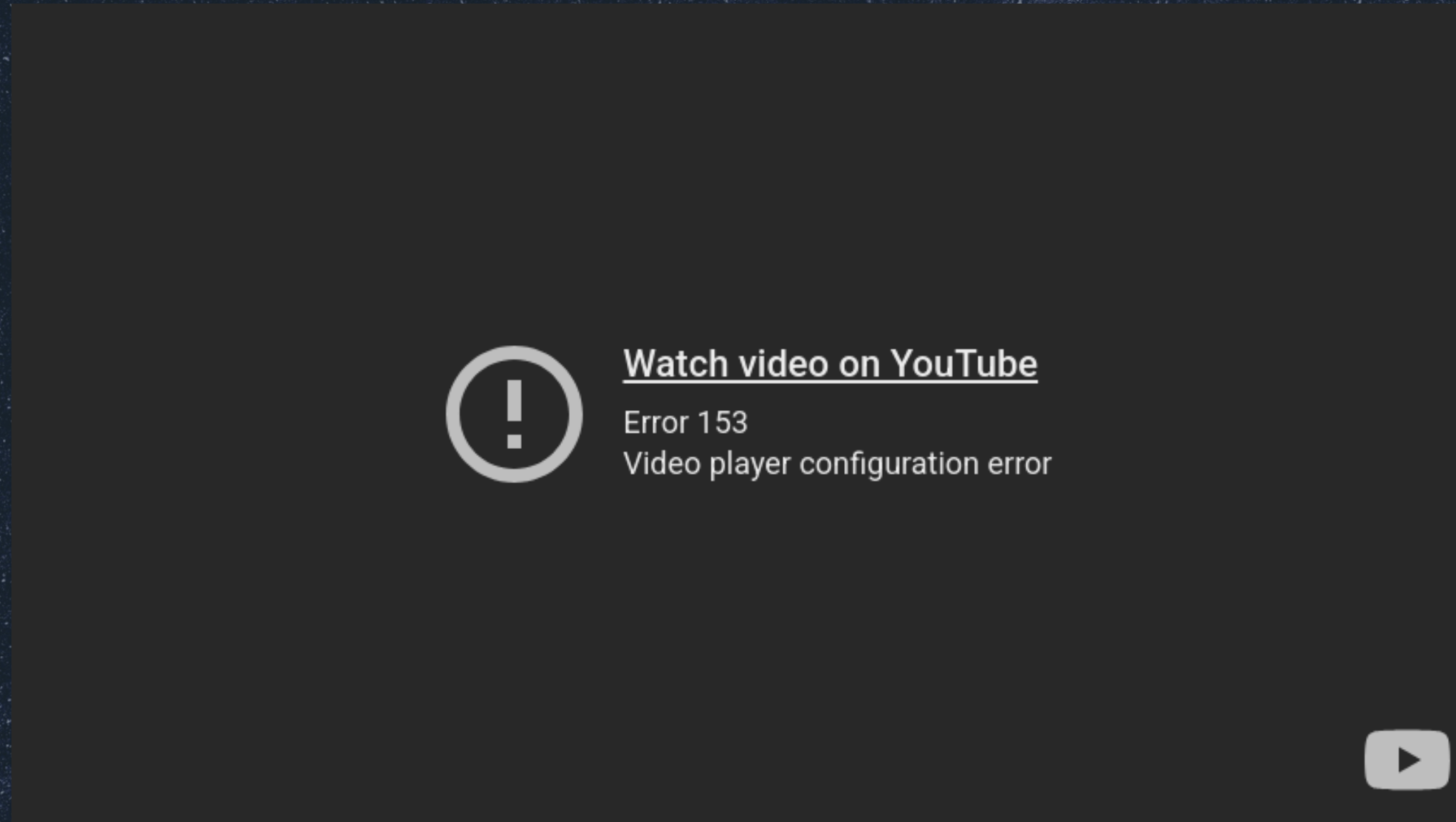
This one is even more popular on the sleep side of youtube than binaural beats. 432 Hz is said to be harmonious with nature, leading to benefits like reduced stress, enhanced clarity, and improved sleep quality. 440 Hz is the modern standard tuning frequency, but in many wellness communities 432 is said to be preferred.

“I feel more productive” (from 0 – not at all to 10 – very much)		Differences in productivity	
	Mean (SD), Range	Group Comparison	F, p
440 Hz Group 1	6.857 (1.556), 3-10	Group 1 vs Group 3 (control)	F=20.358, p=0.000**
432 Hz Group 2	6.286 (2.462), 1-10	Group 1 vs Group 2	F=6.601, p=0.304
Group 3 (control)	4.11 (2.913), 0-8	Group 2 vs Group 3 (control)	F=2.508, p=0.004**

One study done on nurses during the pandemic. Nurses did not know which track they were listening to

48 MILLION VIEWS?!?!

May all The overthinking be remove by love peace of mind love and clarity



I used to smoke 5 packs of cigarettes a day and this audio helped me quit

This is what Heaven will sound like

I don't know If someone is reading this, but if you are: You are amazing and beautiful! I believe in you!

Another 432 Hz video... is it just as relaxing? You decide



[Watch video on YouTube](#)

Error 153

Video player configuration error



References:

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC9163611/>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC6924256/>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC8257833/>
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- <https://pmc.ncbi.nlm.nih.gov/articles/PMC6165862/>
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- https://www.researchgate.net/publication/363389691_Validation_Study_of_the_Richards-Campbell_Sleep_Questionnaire_in_Patients_with_Acute_Stroke/download
- <https://www.theguardian.com/lifeandstyle/2020/oct/18/white-noise-as-sleep-aid-may-do-more-harm-than-good-say-scientists>